

✓ Key Cost Driver Analysis: Shipping Cost

Overview

This project analyzes key factors influencing shipping costs in a dynamic supply chain environment using exploratory data analysis (EDA) and correlation techniques.

The goal is to evaluate whether commonly assumed operational variables such as fuel consumption, loading/unloading time, and congestion levels significantly impact shipping cost.

Dataset

Source: Kaggle – Logistics and Supply Chain Dataset

Total Records: ~32,000

Features: 26 variables including:

Fuel consumption rate

Traffic congestion level

Warehouse inventory level

Loading/unloading time

Lead time

Customs clearance time

Shipping costs

Objective

To identify and quantify relationships between shipping costs and operational, environmental, and logistical factors.

Analysis Performed

Data Extraction & Preparation

1. Dataset loaded using pandas
2. Selected relevant variables for cost analysis: shipping_costs:
 - shipping_costs
 - fuel_consumption_rate
 - loading_unloading_time

```
## Extract - install kaggle
```

```
!pip install kagglehub
```

```
Requirement already satisfied: kagglehub in /usr/local/lib/python3.12/dis
Requirement already satisfied: kagglesdk<1.0,>=0.1.14 in /usr/local/lib/p
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dis
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: protobuf in /usr/local/lib/python3.12/dist
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/pytho
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/pytho
```

```
#Download Dataset from kaggle
```

```
import kagglehub
```

```
path = kagglehub.dataset_download("datasetengineer/logistics-and-supply-
print("dataset path:", path)
```

```
Using Colab cache for faster access to the 'logistics-and-supply-chain-da
dataset path: /kaggle/input/logistics-and-supply-chain-dataset
```

```
import os
```

```
print(os.listdir(path))
```

```
['dynamic_supply_chain_logistics_dataset.csv']
```

```
import pandas as pd
```

```
file_path = os.path.join(path, "dynamic_supply_chain_logistics_dataset.c
```

```
df = pd.read_csv(file_path)
```

```
df.head(5)
```

	timestamp	vehicle_gps_latitude	vehicle_gps_longitude	fuel_consumption
0	2021-01-01 00:00:00	40.375568	-77.014318	5
1	2021-01-01 01:00:00	33.507818	-117.036902	5
2	2021-01-01 02:00:00	30.020640	-75.269224	5
3	2021-01-01 03:00:00	36.649223	-70.190529	8
4	2021-01-01 04:00:00	30.001279	-70.012195	5

5 rows × 26 columns

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 32065 entries, 0 to 32064
Data columns (total 26 columns):
```

#	Column	Non-Null Count	Dtype
0	timestamp	32065 non-null	object
1	vehicle_gps_latitude	32065 non-null	float64
2	vehicle_gps_longitude	32065 non-null	float64
3	fuel_consumption_rate	32065 non-null	float64
4	eta_variation_hours	32065 non-null	float64
5	traffic_congestion_level	32065 non-null	float64
6	warehouse_inventory_level	32065 non-null	float64
7	loading_unloading_time	32065 non-null	float64
8	handling_equipment_availability	32065 non-null	float64
9	order_fulfillment_status	32065 non-null	float64
10	weather_condition_severity	32065 non-null	float64
11	port_congestion_level	32065 non-null	float64
12	shipping_costs	32065 non-null	float64
13	supplier_reliability_score	32065 non-null	float64
14	lead_time_days	32065 non-null	float64
15	historical_demand	32065 non-null	float64
16	iot_temperature	32065 non-null	float64
17	cargo_condition_status	32065 non-null	float64
18	route_risk_level	32065 non-null	float64
19	customs_clearance_time	32065 non-null	float64
20	driver_behavior_score	32065 non-null	float64
21	fatigue_monitoring_score	32065 non-null	float64
22	disruption_likelihoood_score	32065 non-null	float64
23	delay_probability	32065 non-null	float64
24	risk_classification	32065 non-null	object
25	delivery_time_deviation	32065 non-null	float64

dtypes: float64(24), object(2)

memory usage: 6.4+ MB

```

selected_columns = [
    'shipping_costs',
    'fuel_consumption_rate',
    'traffic_congestion_level',
    'warehouse_inventory_level',
    'loading_unloading_time',
    'lead_time_days',
    'customs_clearance_time',
]

corr = df[selected_columns].corr()
print(corr['shipping_costs'].sort_values(ascending=False))

```

```

shipping_costs          1.000000
fuel_consumption_rate    0.005036
loading_unloading_time   0.003086
traffic_congestion_level -0.000031
customs_clearance_time   -0.000841
warehouse_inventory_level -0.005494
lead_time_days           -0.007044
Name: shipping_costs, dtype: float64

```

```

cost_analysis = df[['timestamp', 'shipping_costs', 'fuel_consumption_rate',
                    'nlargest(10, 'fuel_consumption_rate')'].round(2)

```

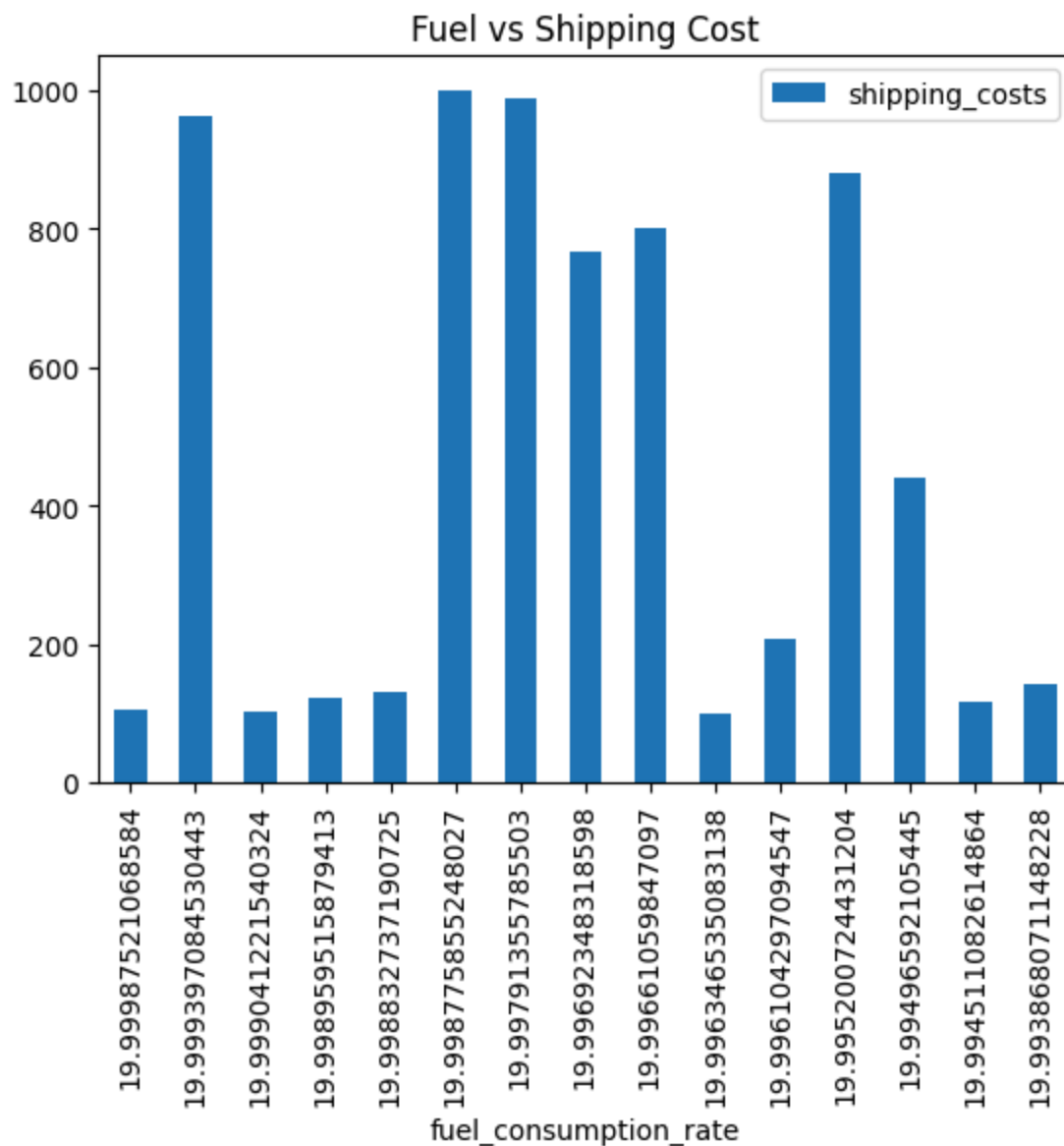
```
cost_analysis
```

	timestamp	shipping_costs	fuel_consumption_rate
9132	2022-01-16 12:00:00	105.47	20.0
5800	2021-08-30 16:00:00	964.05	20.0
4474	2021-07-06 10:00:00	101.13	20.0
28062	2024-03-15 06:00:00	122.54	20.0
3853	2021-06-10 13:00:00	130.45	20.0
18849	2023-02-25 09:00:00	999.99	20.0
19547	2023-03-26 11:00:00	989.56	20.0
21679	2023-06-23 07:00:00	767.48	20.0
4721	2021-07-16 17:00:00	801.81	20.0
17312	2022-12-23 08:00:00	100.52	20.0

```
import matplotlib.pyplot as plt

Cost_analysis.plot( x='fuel_consumption_rate', y='shipping_costs',kind='

plt.title('Fuel vs Shipping Cost')
plt.show()
```

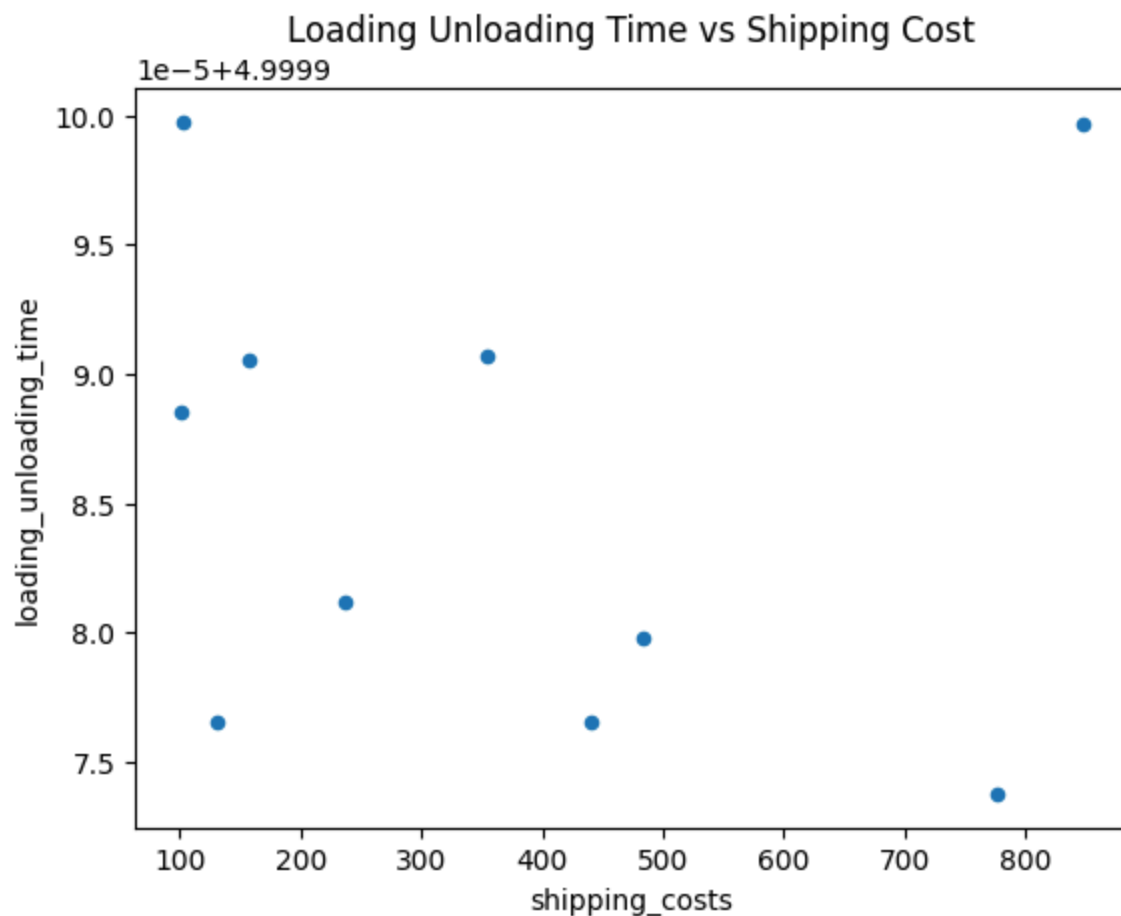


```
cost_analysis_based_on_loading = df[['shipping_costs', 'loading_unloadin
    .nlargest(10, 'loading_unloading_time', )

cost_analysis_based_on_loading
```

	shipping_costs	loading_unloading_time
10406	101.800275	5.000000
15111	846.924976	5.000000
12915	354.520910	4.999991
26685	157.842239	4.999991
14742	100.380650	4.999988
845	236.783191	4.999981
13030	483.387509	4.999980
18210	440.922654	4.999977
20586	131.457342	4.999977
18386	776.372801	4.999974

```
cost_analysis_based_on_loading.plot (x = 'shipping_costs', y = 'loading_
plt.title('Loading Unloading Time vs Shipping Cost')
plt.show())
```



Conclusion

In conclusion, the analysis demonstrates that commonly assumed operational factors like fuel consumption and loading/unloading time do not significantly explain variations in shipping costs in this dataset. This highlights the importance of using more advanced analytical approaches and incorporating additional variables to better understand cost behavior in supply chain operations.

Thanks!